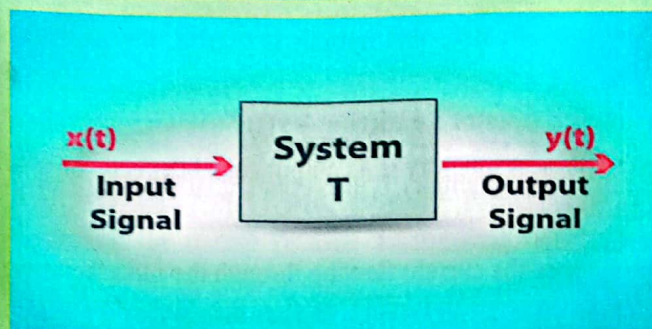


AL-Ayen Iraqi University
Engineering Collage / AI Engineering Department

Signals and Systems

Lecture 4 : Systems



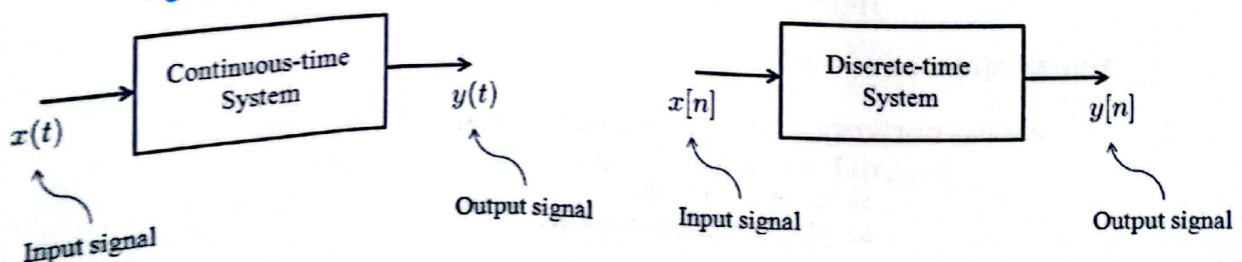
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إشارات ونظم نهج معالجة
ذكاء ثالثة

Introduction to Systems

- Systems are used to process signals to allow modification or extraction of additional information from the signal.
- A system may consist of physical components (hardware realization) or an algorithm (operator) that computes the output signal from the input signal.
- A physical system consists of inter-connected components which are characterized by their input-output relationships.

Figure 2.1: Continuous-time and discrete-time systems: Here H & T are operators.



Properties of systems: (Classification of systems):

1 Static (Memoryless) & Dynamic (with memory):

Static: A system is static if the output at time t (or n) depends only on the input at time t (or n).

Examples:

1. $y(t) = (2x(t) - x^2(t))^2$ is memoryless, because $y(t)$ depends on $x(t)$ only. There is no $x(t-1)$, or $x(t+1)$ terms, for example.
2. $y[n] = x^2[n]$ is memoryless. In fact, this system is passing the input to output directly, without any processing.
3. Current flowing through a resistor i.e., $i(t) = \frac{1}{R} v(t)$

Dynamic: A system is said to possess memory if its output signal depends on past or future values of input.

Example:

1. Inductor and capacitor, since the current flowing through the inductor at time ' t ' depends on the all past values of the voltage $v(t)$ i.e., $i(t) = \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau$ and $v(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau$
2. The moving average system given by $y(n) = \frac{1}{3} (x[n] + x[n-1] + x[n-2])$

2 Stable & unstable system:

- A system is said to be bounded-input, bounded-output (BIBO) stable if and only if every bounded input results in a bounded output, otherwise it is said to be unstable.
- If for $|x(t)| \leq M_x < \infty$ for all t , output is $|y(t)| \leq M_y < \infty$ for all t ; where M_x & M_y are some finite positive number.

Example: 1. $y(t) = x(t-3)$ is a stable system.
2. $y(t) = t x(t)$ is an unstable system.
3. $y[n] = e^{x[n]}$ is a stable system.

Assume that $|x(n)| \leq M_x < \infty$, for all ' t ',
 $y[n] = e^{x[n]} = e^{M_x} = \text{finite} \rightarrow \text{Stable}$

4. $y[n] = r^n x[n]$, where $r > 1$

Assume that $|x(n)| \leq M_x < \infty$, for all ' t ', then
 $|y[n]| = |r^n x[n]| = |r^n| |x[n]|$
as ' $n \rightarrow \infty$ ' $|r^n| \rightarrow \infty$
so $y[n] \rightarrow \infty$ hence unstable.

3 Causal and non-Causal system:

Causal: A system is said to be causal if the present value of output signal depends only on the present or past values of the input signal. A causal system is also known as physical or non-anticipative system.

Example: 1. The moving average system given by $y(n) = \frac{1}{3} (x[n] + x[n-1] + x[n-2])$

2. $y(t) = x(t) \cos 6t$

Note: i) Any practical system that operates in real time must necessarily be causal.
ii) All static systems are causal.

Non-Causal: A system is said to be non-causal if the present value of output signal depends on one or more future values of the input signal.

Example: 1. The moving average system given by $y(n) = \frac{1}{3} (x[n] + x[n-1] + x[n+2])$

4 Time invariant and time variant system:

Time invariant: A system is time-invariant if a time-shift of the input signal results in the same time-shift of the output signal.

That is, if

$$x(t) \rightarrow y(t),$$

then the system is time-invariant if

$$x(t - t_0) \rightarrow y(t - t_0), \text{ for any } t_0.$$

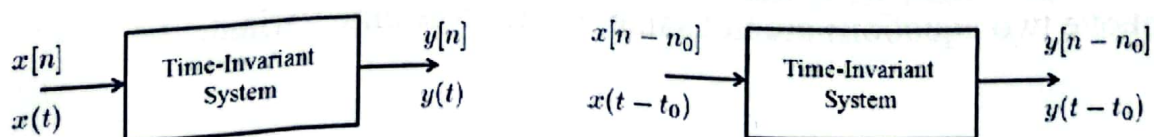


Figure 2.2: Illustration of a time-invariant system.

Example 1.

The system $y(t) = \sin[x(t)]$ is time-invariant

Proof. Let us consider a time-shifted signal $x_1(t) = x(t - t_0)$. Correspondingly, we let $y_1(t)$ be the output of $x_1(t)$. Therefore,

$$y_1(t) = \sin[x_1(t)] = \sin[x(t - t_0)].$$

Now, we have to check whether $y_1(t) = y(t - t_0)$. To show this, we note that

$$y(t - t_0) = \sin[x(t - t_0)],$$

which is the same as $y_1(t)$. Therefore, the system is time-invariant. $\square$

Time variant: A system is time-variant if its input-output characteristic changes with time.

Example 2:

The system $y[n] = nx[n]$ is time-variant.

Proof: Output for a time shifted input is

$$y[n] \mid x(n-k) = nx(n-k)$$

then the same time shifted output is

$$y(n-k) = (n-k)x(n-k)$$

the above two equations are not same. Hence it is time variant.

4 Linear and non-linear system:

Linear system: A system is said to be linear if it satisfies two properties i.e.; superposition & homogeneity.

Superposition: It states that the response of the system to a weighted sum of signals be equal to the corresponding weighted sum of responses (Outputs of the system to each of the individual input signal).

For an input $x(t) = x_1(t)$, the output $y(t) = y_1(t)$

and input $x(t) = x_2(t)$, the output $y(t) = y_2(t)$

then, the system is linear if & only if

$$\mathcal{T}[a_1x_1(t) + a_2x_2(t)] = a_1\mathcal{T}[x_1(t)] + a_2\mathcal{T}[x_2(t)]$$

Homogeneity: If the input $x(t)$ is scaled by a constant factor 'a', then the output $y(t)$ is also scaled by exactly the same constant factor 'a'.

For an input $x(t) \rightarrow$ output $y(t)$

and input $x_1(t) = ax(t) \rightarrow$ output $y_1(t) = ay(t)$

Example 1:

The system $y(t) = 2\pi x(t)$ is linear. To see this, let's consider a signal

$$x(t) = ax_1(t) + bx_2(t),$$

where $y_1(t) = 2\pi x_1(t)$ and $y_2(t) = 2\pi x_2(t)$. Then

$$\begin{aligned} ay_1(t) + by_2(t) &= a(2\pi x_1(t)) + b(2\pi x_2(t)) \\ &= 2\pi [ax_1(t) + bx_2(t)] = 2\pi x(t) = y(t). \end{aligned}$$

Example 2.

The system $y[n] = (x[2n])^2$ is not linear. To see this, let's consider the signal

$$x[n] = ax_1[n] + bx_2[n],$$

where $y_1[n] = (x_1[2n])^2$ and $y_2[n] = (x_2[2n])^2$. We want to see whether $y[n] = ay_1[n] + by_2[n]$. It holds that

$$ay_1[n] + by_2[n] = a(x_1[2n])^2 + b(x_2[2n])^2.$$

However,

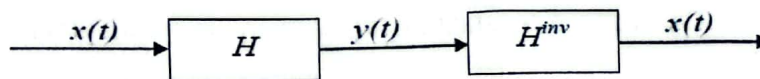
$$y[n] = (x[2n])^2 = (ax_1[2n] + bx_2[2n])^2 = a^2(x_1[2n])^2 + b^2(x_2[2n])^2 + 2abx_1[n]x_2[n].$$

5 Invertible and non-invertible system:

A system is said to be invertible if the input of the system can be recovered from the output.

Let the set of operations needed to recover the input represents the second system which is connected in cascade with the given system such that the output signal of the second system is equal to the input applied to the given system.

Let $H \rightarrow$ the continuous time system
 $x(t) \rightarrow$ input signal to the system
 $y(t) \rightarrow$ output signal of the system
 $H^{inv} \rightarrow$ the second continuous time system



The output signal of the second system is given by

$$\begin{aligned} H^{inv}\{y(t)\} &= H^{inv}\{Hx(t)\} \\ &= H^{inv}H\{x(t)\} \end{aligned}$$

For the output signal to equal to the original input, we require that

$$H H^{inv} = I$$

Where ' I ' denotes the identity operator.

The system whose output is equal to the input is an identity system. The operator H^{inv} must satisfy the above condition for H to be an invertible system. Cascading a system, with its inverse system, result in an identity system.

Example:

An inductor is described by the relation

$$y(t) = \frac{1}{L} \int_{-\infty}^t x(\tau) d\tau \text{ is an invertible system}$$

because, by rearranging terms, we get

$$x(t) = L \frac{d}{dt} y(t),$$

which is the inversion formula.

Note:i) A system is not invertible unless distinct inputs applied to the system produce distinct outputs.

ii) There must be a one to one mapping between input and output signal for system to be invertible.

Non-invertible System: When several different inputs results in the same output, it is impossible to obtain the input from output. Such system is called a non-invertible system.

Example: A square-law system described by the input output relation

$$y(t) = x^2(t), \text{ is non-invertible,}$$

because distinct inputs $x(t)$ & $-x(t)$ produce the same output $y(t)$ [not distinct output].

Linear –time convolution system (LTI)

Linear time invariant (LTI) systems are good models for many real-life systems, and they have properties that lead to a very powerful and effective theory for analyzing their behavior. The LTI systems can be studied through its characteristic function, called the impulse response. Further, any arbitrary input signal can be decomposed and represented as a weighted sum of unit sample sequences. As a consequence of the linearity and time invariance properties of the system, the response of the system to any arbitrary input signal can be expressed in terms of the unit sample response of the system. The general form of the expression that relates the unit sample response of the system and the arbitrary input signal to the output signal, called the convolution sum, is also derived.

Resolution of a Discrete-time signal into impulses:

Any arbitrary sequence $x(n)$ can be represented in terms of delayed and scaled impulse sequence $\delta(n)$. Let $x(n)$ is an infinite sequence as shown in figure below.

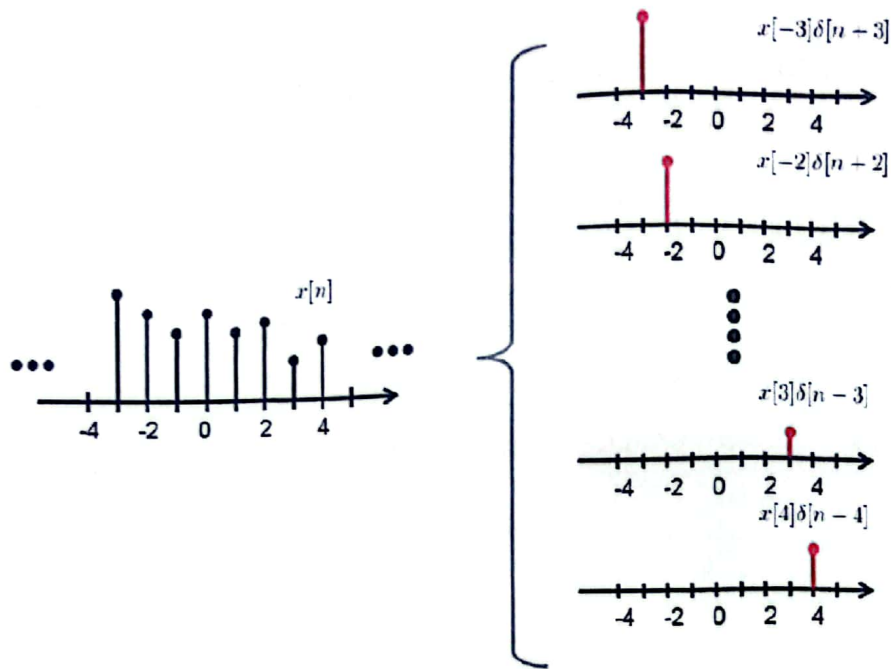


Figure 1.13: Representing of a signal $x[n]$ using a train of impulses $\delta[n - k]$.

The sample $x(0)$ can be obtained by multiplying $x(0)$, the magnitude, with unit impulse $\delta(n)$

$$\text{i.e., } x[n] \delta[n] = \begin{cases} x(0), & n=0 \\ 0, & n \neq 0 \end{cases}$$

Similarly, the sample $x(-3)$ can be obtained as shown in the figure.

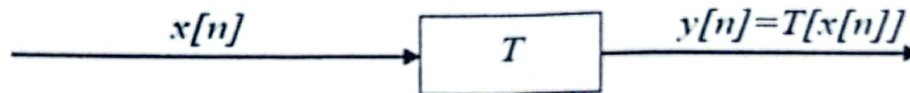
$$\text{i.e., } x[-3] \delta[n+3] = \begin{cases} x(-3), & n=-3 \\ 0, & n \neq -3 \end{cases}$$

In the same way we can get the sequence $x[n]$ by summing all the shifted and scaled impulse function

$$\begin{aligned} \text{i.e., } x[n] &= \dots x[-3] \delta[n+3] + x[-2] \delta[n+2] + \dots + x[0] \delta[n] + \dots + x[4] \delta[n-4] \dots \\ &= \sum_{k=-\infty}^{\infty} x(k) \delta(n - k) \end{aligned}$$

Impulse response and convolution sum:

Impulse response: A discrete-time system performs an operation on an input signal based on predefined criteria to produce a modified output signal. The input signal $x[n]$ is the system excitation, and $y[n]$ is the system response. The transform operation is shown in the figure below.



If the input to the system is the unit impulse i.e., $x[n] = \delta[n]$, then the output of the system is known as impulse response represented by $h[n]$ where

$$h[n] = T[\delta[n]]$$

Response of LTI system to arbitrary inputs: The convolution sum

From the above discussion, we get the response of an LTI system to an unit impulse as the impulse response $h[n]$ i.e.,

$$\delta[n] \longrightarrow h[n]$$

$$\delta[n-k] \longrightarrow h[n-k], \text{ by time invariant property}$$

$$x(k)\delta[n-k] \longrightarrow x(k)h[n-k], \text{ by homogeneity principle}$$

$$\sum_{k=-\infty}^{\infty} x(k)\delta[n-k] \longrightarrow \sum_{k=-\infty}^{\infty} x(k)h[n-k], \text{ by super position}$$

As we know the arbitrary input signal is a weighted sum of impulse, the LHS = $x[n]$ having a response in RHS = $y[n]$ known as convolution summation.

$$\text{i.e., } x[n] \longrightarrow y[n]$$

In other words, given a signal $x[n]$ and the impulse response of an LTI system $h[n]$, the convolution between $x[n]$ and $h[n]$ is defined as

$$y[n] = \sum_{k=-\infty}^{\infty} x(k)h[n-k]$$

We denote convolution as $y[n] = x[n] * h[n]$.

- Equivalent form: Letting $m = n - k$, we can show that

$$\sum_{k=-\infty}^{\infty} x(k)h[n-k] = \sum_{m=-\infty}^{\infty} x(n-m)h[m] = \sum_{k=-\infty}^{\infty} x[n-k]h[k]$$

Properties of convolution:

The following “standard” properties can be proved easily:

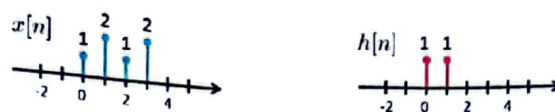
1. Commutative: $x[n] * h[n] = h[n] * x[n]$
2. Associative: $x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$
3. Distributive: $x[n] * (h_1[n] + h_2[n]) = (x[n] * h_1[n]) + (x[n] * h_2[n])$

How to Evaluate Convolution?

To evaluate convolution, there are four basic steps:

1. Fold
2. Shift
3. Multiply
4. Summation

Example 1: Consider the signal $x[n]$ and the impulse response $h[n]$ shown below.



Let's compute the output $y[n]$ one by one. First, consider $y[0]$:

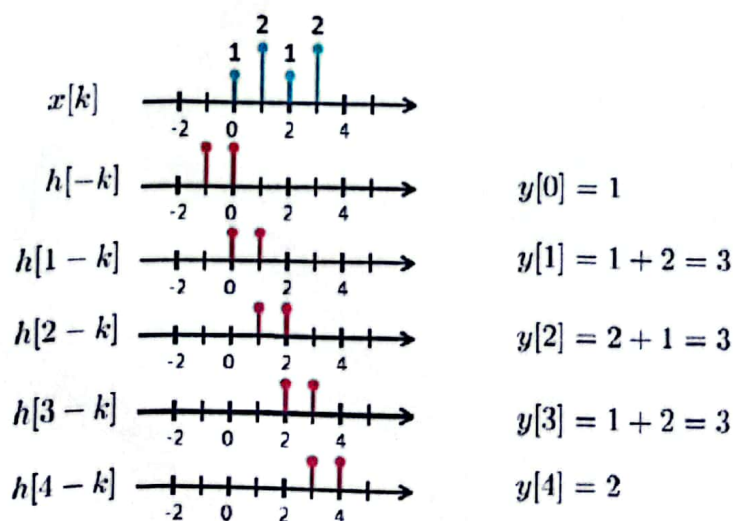
$$y[0] = \sum_{k=-\infty}^{\infty} x(k)h[0-k] = \sum_{k=-\infty}^{\infty} x(k)h[-k] = 1$$

Note that $h[-k]$ is the flipped version of $h[k]$, and $\sum_{k=-\infty}^{\infty} x(k)h[-k] = 1$ is the multiply-add between $x[k]$ and $h[-k]$.

To calculate $y[1]$, we flip $h[k]$ to get $h[-k]$, shift $h[-k]$ to get $h[1-k]$, and multiply-add to get $\sum_{k=-\infty}^{\infty} x(k)h[1-k]$. Therefore

$$y[1] = \sum_{k=-\infty}^{\infty} x(k)h[1-k] = \sum_{k=-\infty}^{\infty} x(k)h[1-k] = 1 \times 1 + 2 \times 1 = 3$$

The calculation is shown in the figure below.



Assignment

1. Classify each of the following systems as linear or nonlinear:

a) $y(t) = 4 + 3x(t)$

b) $y(t) = [x(t)]^2$

c) $y(t) = x(t) + (1 - 2x(t))$

2. Determine whether each system is time-invariant or time-variant:

a) $y(t) = x(t - 3)$

b) $y(t) = t \cdot x(t)$

3. Identify whether each system is causal or non-causal:

a) $y(t) = x(t - 1) + (2 - x(t))$

b) $y(t) = x(t + 2)$

4. Check if the following systems are memoryless or dynamic:

a) $y(t) = 5x(t)$

b) $y(t) = 1 - x(t)$

THANK YOU